
O-RAN Work Group 11 (Security work group)

Study on Post-Quantum Cryptography

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Foreword

This Technical Report (TR) has been produced by WG11 of the O-RAN ALLIANCE.

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Introduction

The increasing sophistication of quantum computing poses a significant threat to the security of classical asymmetric cryptographic algorithms currently used in O-RAN networks. As quantum computers become more powerful, they will be capable of breaking asymmetric classical encryption algorithms, potentially compromising the confidentiality, integrity, and authenticity of sensitive information exchanged between network entities. This vulnerability has far-reaching implications for the security and trustworthiness of O-RAN networks, which are designed to provide secure and reliable communication services for critical infrastructure and applications.

The migration to Quantum Safe Cryptography (QSC) is essential to ensure the long-term security and resilience of O-RAN networks against potential quantum computer-based attacks. QSC refers to cryptographic techniques and algorithms that are resistant to attacks by both classical and quantum computers. By migrating to QSC, O-RAN networks can maintain their security posture and prevent potential disruptions to services and operations.

This technical report contains a cryptographic inventory, covering all cryptographic algorithms employed in O-RAN specification at this time.

1 Scope

The present document presents the Post-Quantum Cryptography study as it applies to O-RAN architectural elements.

2 References

2.1 Normative references

Not applicable.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”.
- [i.2] NIST Post-Quantum Cryptography, call for papers; <https://csrc.nist.gov/Projects/post-quantum-cryptography/post-quantum-cryptography-standardization/Call-for-Proposals>
- [i.3] FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard; <https://csrc.nist.gov/pubs/fips/203/final>
- [i.4] FIPS 204: Module-Lattice-Based Digital Signature Standard; <https://csrc.nist.gov/pubs/fips/204/final>
- [i.5] FIPS 205: Stateless Hash-Based Digital Signature Standard; <https://csrc.nist.gov/pubs/fips/205/final>
- [i.6] NCSC whitepaper: Next steps in preparing for post-quantum cryptography; <https://www.ncsc.gov.uk/whitepaper/next-steps-preparing-for-post-quantum-cryptography>
- [i.7] ETSI TR 103 619 V1.1.1 (2020-07), “Migration strategies and recommendations to Quantum Safe schemes”
- [i.8] ETSI TC CYBER, ETSI QSC, “Quantum Safe Cryptographic Protocol Inventory”, CYBERQSC(23)032007r2_QSC_Protocol_Inventory_LSOOut.pdf
- [i.9] GSMA, “LS regarding the publication of the Post Quantum Cryptography – Guidelines for Telecom Use Cases document in Feb 24”
- [i.10] GSMA, “Post Quantum Cryptography – Guidelines for Telecom Use Cases”; <https://www.gsma.com/newsroom/wp-content/uploads/PQ.03-Post-Quantum-Cryptography-Guidelines-for-Telecom-Use-v1.0.pdf>
- [i.11] NIST, Post-Quantum Cryptography; <https://www.nist.gov/programs-projects/post-quantum-cryptography>
- [i.12] O-RAN ALLIANCE TS “O-RAN WG4 Management Plane Specification”
- [i.13] O-RAN ALLIANCE TS “O-RAN WG10 O-RAN O1 Interface Specification”

- [i.14] O-RAN ALLIANCE TS “O-RAN WG11 Security Protocols Specifications”
- [i.15] IETF RFC 5656: “Elliptic Curve Algorithm Integration in the Secure Shell Transport Layer”
- [i.16] IETF RFC 8709: “Ed25519 and Ed448 Public Key Algorithms for the Secure Shell (SSH) Protocol”
- [i.17] IETF RFC 5647: “AES Galois Counter Mode for the Secure Shell Transport Layer Protocol”
- [i.18] IETF RFC 4344: “The Secure Shell (SSH) Transport Layer Encryption Modes”
- [i.19] IETF RFC 4419: “Diffie-Hellman Group Exchange for the Secure Shell (SSH) Transport Layer Protocol”
- [i.20] IETF RFC 8270. “Increase the Secure Shell Minimum Recommended Diffie-Hellman Modulus Size to 2048 Bits”
- [i.21] IETF RFC 8731: “Secure Shell (SSH) Key Exchange Method Using Curve25519 and Curve448”
- [i.22] IETF RFC 6668: “SHA-2 Data Integrity Verification for the Secure Shell (SSH) Transport Layer Protocol”
- [i.23] IETF RFC 4418: “UMAC: Message Authentication Code using Universal Hashing”
- [i.24] O-RAN ALLIANCE TS "A1 interface: Transport Protocol"
- [i.25] O-RAN ALLIANCE TS "Transport Protocols for R1 Services"
- [i.26] O-RAN ALLIANCE TS "Y1 interface: General Aspects and Principles"
- [i.27] IETF RFC 5116: "An Interface and Algorithms for Authenticated Encryption"
- [i.28] IETF RFC 8422: "Elliptic Curve Cryptography (ECC) Cipher Suites for Transport Layer Security (TLS) Versions 1.2 and Earlier"
- [i.29] IETF RFC 7919: "Negotiated Finite Field Diffie-Hellman Ephemeral Parameters for Transport Layer Security (TLS) "
- [i.30] IETF RFC 6234: "US Secure Hash Algorithms (SHA and SHA-based HMAC and HKDF) "
- [i.31] IETF RFC 8446: "The Transport Layer Security (TLS) Protocol Version 1.3"
- [i.32] IETF RFC 8439: "ChaCha20 and Poly1305 for IETF Protocols"
- [i.33] BSI TR-02102-2: "Cryptographic Mechanisms: Recommendations and Key Lengths: Use of Transport Layer Security (TLS) "
- [i.34] NIST SP 800-38D: "Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC"
- [i.35] IETF RFC 8032: "Edwards-Curve Digital Signature Algorithm (EdDSA)"
- [i.36] IETF RFC 7518: "JSON Web Algorithms (JWA)"
- [i.37] O-RAN ALLIANCE TS "O-RAN Near-RT RIC Architecture"
- [i.38] O-RAN ALLIANCE TS "O-RAN Acceleration Abstraction Layer General Aspects and Principles"
- [i.39] O-RAN ALLIANCE TS “O-RAN WG6 O2 Interface General Aspects and Principles”
- [i.40] IETF RFC 7516: "JSON Web Encryption (JWE)"
- [i.41] IEEE Std 802.1AE-2018, IEEE Standard for Local and metropolitan area networks — Media Access Control (MAC) Security, IEEE, 2018
- [i.42] IEEE 802.1X-2020, “IEEE Standard for Local and Metropolitan Area Networks--Port-Based Network Access Control”, IEEE, 2020-02-28

- [i.43] O-RAN ALLIANCE TS “O-RAN WG11 Study on Security for Fronthaul CUS-Plane”
- [i.44] IETF RFC 4493: “The AES-CMAC Algorithm”
- [i.45] IETF RFC 5216: “The EAP-TLS Authentication Protocol”
- [i.46] IETF RFC 9190: “EAP-TLS 1.3: Using the Extensible Authentication Protocol with TLS 1.3”
- [i.47] IEEE Std 1588 – 2019, IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems, IEEE, 2019
- [i.48] IETF RFC 2410: The NULL Encryption Algorithm and Its Use with IPsec"
- [i.49] IETF RFC 2404: The Use of HMAC-SHA-1-96 within ESP and AH"
- [i.50] IETF RFC 3526: More Modular Exponential (MODP) Diffie-Hellman groups for Internet Key Exchange (IKE)"
- [i.51] IETF RFC 3566: The AES-XCBC-MAC-96 Algorithm and Its Use with IPsec"
- [i.52] IETF RFC 3602: The AES-CBC Cipher Algorithm and Its Use with IPsec"
- [i.53] IETF RFC 4106: The Use of Galois/Counter Mode (GCM) in IPsec Encapsulating Security Payload (ESP)"
- [i.54] IETF RFC 4309: Using Advanced Encryption Standard (AES) CCM Mode with IPsec Encapsulating Security Payload (ESP)"
- [i.55] IETF RFC 4543: The Use of Galois Message Authentication Code (GMAC) in IPsec ESP and AH"
- [i.56] IETF RFC 4868: Using HMAC-SHA-256, HMAC-SHA-384, and HMAC-SHA-512 with IPsec"
- [i.57] IETF RFC 5114: Additional Diffie-Hellman Groups for Use with IETF Standards"
- [i.58] IETF RFC 5282: Using Authenticated Encryption Algorithms with the Encrypted Payload of the Internet Key Exchange version 2 (IKEv2) Protocol"
- [i.59] IETF RFC 7296: Internet Key Exchange Protocol Version 2 (IKEv2)"
- [i.60] IETF RFC 7634: ChaCha20, Poly1305, and Their Use in the Internet Key Exchange Protocol (IKE) and IPsec"
- [i.61] IETF RFC 8221: Cryptographic Algorithm Implementation Requirements and Usage Guidance for Encapsulating Security Payload (ESP) and Authentication Header (AH)"
- [i.62] IETF RFC 8247: Algorithm Implementation Requirements and Usage Guidance for the Internet Key Exchange Protocol Version 2 (IKEv2)
- [i.63] O-RAN ALLIANCE TS “O-RAN WG3 E2 General Aspects and Principles (E2GAP)”
- [i.64] O-RAN ALLIANCE TS: "O-RAN Architecture Description"
- [i.65] IETF RFC 7519: "JSON Web Token (JWT)"
- [i.66] IETF RFC 6749: "The OAuth 2.0 Authorization Framework"

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in [i.1] and the following apply:

Post-Quantum Cryptography: An area of cryptography that researches and advances the use of quantum-resistant primitives, with the goal of keeping existing public key infrastructure intact in a future era of quantum computing. Intended to be secure against both quantum and classical computers and deployable without drastic changes to existing communication protocols and networks. [i.11]

Quantum Safe Cryptography: see Post-Quantum Cryptography

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

CUS-Plane	Control, User and Synchronization Plane
DTLS	Datagram Transport Layer Security
ETM	Encrypt-then-MAC
FH	Fronthaul
HMAC	Hash-based message authentication code
JSON	JavaScript Object Notation
JWA	JSON Web Algorithms
JWE	JSON Web Encryption
JWS	JSON Web Signature
JWT	JSON Web Token
MACsec	Media Access Control security
MGF	Mask generation function
MKA	MACsec Key Agreement
PQC	Post-Quantum Cryptography
PTP	Precision Time Protocol
QSC	Quantum Safe Cryptography
SSH	Secure Shell
TLS	Transport Layer Security

4 Post-Quantum Cryptography background

NIST has initiated the process of the Post Quantum Cryptography (PQC) standardization in 2017 [i.2]. On 5th Jul 2022, they announced the standard candidates for public-key encryption:

- CRYSTALS-Kyber,
- for digital signature: CRYSTALS-Dilithium, FALCON and SPHINCS+.

On 13th Aug 2024, NIST published FIPS 203 [i.3], FIPS 204 [i.4] and FIPS 205 [i.5], specifying algorithms derived from CRYSTALS-Dilithium, CRYSTALS-KYBER and SPHINCS+: ML-KEM, ML-DSA and SLH-DSA. A future FIPS 206 is planned to specify FN-DSA, based on FALCON.

Further input from NIST:

“Quantum computers would completely break many public-key cryptosystems, including RSA, DSA, and elliptic curve cryptosystems. These cryptosystems are used to implement digital signatures and key establishment and play a crucial role in ensuring the confidentiality and authenticity of communications on the Internet and other networks.” [i.2]

Further input from NCSC:

“In contrast with PKC, the security of symmetric cryptography is not significantly impacted by quantum computers, and existing symmetric algorithms with at least 128-bit keys (such as AES) can continue to be used. The security of hash functions such as SHA-256 is also not significantly affected, and secure hash functions can also continue to be used.” [i.6]

The ETSI QSC group has developed a document outlining a Quantum Safe Cryptography migration strategy and recommendations for transitioning to quantum-safe schemes [i.7]. A key component of this strategy is the creation of a cryptographic inventory or cryptographic bill of material (CBOM) to support the development of a preparation and PQC migration plan, as well as the execution of the migration.

According to the LS [i.8], the ETSI QSC is seeking additional information, such as a "detailed list of cryptographic algorithms and/or protocol implementations." This list may include "guidance on algorithm exposure, effective key lengths, potential for hybrid deployment, and potential for crypto agility."

In a related development, as noted in another LS [i.9], the GSMA has produced a Guidelines for Telecom Use Cases document [i.10], which includes the development of a ""comprehensive cryptographic inventory"" to some extent.

5 Cryptographic inventory

Table 5-1: Cryptographic inventory

Security Protocol	Traditional Algorithm	Algorithm Development Organisation and document(s)	Algorithm used for	O-RAN Adoption (Interfaces, APIs)	O-RAN specification(s)
TLS 1.2, DTLS 1.2	AES-128-GCM	RFC 5116 [i.27] (NOTE 3)	Cipher suites	OFH m-plane [i.12] O1 [i.13] A1 [i.24] R1 [i.25] Y1 [i.26] O2 [i.39] SMO External Interface [i.64]	SPS [i.14]
	AES-256-GCM				
	secp256r1	RFC 8422 [i.28]	Key exchange		
	secp384r1				
	ffdhe2048	RFC 7919 [i.29]			
	ffdhe3072				
	ffdhe4096				
	ffdhe6144				
	ffdhe8192				
	SHA-256	RFC 6234 [i.30]	Hash functions		
	SHA-384				
	ecdsa	RFC 8422 [i.28]	Signature algorithms		
	rsa_pss_rsae_*	RFC 8446 [i.31]			
	rsa_pkcs1	RFC 8032 [i.35] (NOTE 2)			
TLS 1.3, DTLS 1.3	AES-128-GCM	RFC 5116 [i.27] (NOTE 3)	Cipher suites		
	AES-256-GCM				
	ChaCha20-Poly1305	RFC 8439 [i.32]			
	secp256r1	RFC 8422 [i.28]			

Security Protocol	Traditional Algorithm	Algorithm Development Organisation and document(s)	Algorithm used for	O-RAN Adoption (Interfaces, APIs)	O-RAN specification(s)	
	secp384r1	RFC 6234 [i.30]	Key exchange			
	secp521r1		Hash functions			
	SHA-256					
	SHA-384					
	SHA-512					
	ecdsa_secp256r1_sha256	RFC 8446 [i.31]				Signature algorithms
	ecdsa_secp384r1_sha384					
	ecdsa_secp521r1_sha512					
	rsa_pss_rsae_sha256					
	rsa_pss_rsae_sha384					
	rsa_pss_rsae_sha512					
	rsa_pss_pss_sha256					
	rsa_pss_pss_sha384					
	rsa_pss_pss_sha512					
	ed25519		RFC 8032 [i.35]			
SSH	ecdsa-sha2-nistp256	RFC 5656 [i.15][i.15]	Key agreement	OFH m-plane [i.12] O1 [i.13] SMO External Interface [i.64]	SPS [i.14]	
	ecdsa-sha2-nistp384					
	ecdsa-sha2-nistp521					
	ssh-ed25519	RFC 8709 [i.16]				
	ssh-ed448					
	aes256-gcm	RFC 5647 [i.17]	Symmetric encryption and decryption			
	aes128-gcm					
	aes256-ctr	RFC 4344 [i.18]				
	aes192-ctr					
	aes128-ctr					
	ecdh-sha2-nistp521	RFC 5656 [i.15]	Key exchange			
	ecdh-sha2-nistp384					
	ecdh-sha2-nistp256					
	diffie-hellman-group-exchange-sha256	RFC 4419 [i.19], updated by RFC 8270 [i.20]				
	curve25519-sha256					
	hmac-sha2-512-etm	RFC 6668 [i.22] (NOTE 1)	Message authentication codes			
	hmac-sha2-512					
	hmac-sha2-256-etm					
	hmac-sha2-256					
	umac-128	RFC 4418 [i.23]				
IPsec	ENCR_NULL	RFC 2410 [i.48]	ESP encryption algorithms (RFC 8221 – [i.61])	E2 [i.63]	SPS [i.14]	
	ENCR_AES_CBC	RFC 3602 [i.52]				
	ENCR_AES_CCM_8	RFC 4309 [i.54]				
	ENCR_AES_GCM_16	RFC 4106 [i.53]				
	ENCR_CHACHA20_POLY1305	RFC 7634 [i.60]				
	AUTH_NONE (NOTE 7)	RFC 7296 [i.59], RFC 5282 [i.58]	ESP and AH Authentication Algorithms (RFC 8221 – [i.61])			
	AUTH_HMAC_SHA1_96					
	AUTH_AES_XCBC_96 (NOTE 8)	RFC 3566 [i.51], RFC 7296 [i.59]				
	AUTH_AES_128_GMAC	RFC 4543 [i.55]				
	AUTH_AES_256_GMAC	RFC 4543 [i.55]				
	AUTH_HMAC_SHA2_256_128	RFC 4868 [i.56]				
	AUTH_HMAC_SHA2_512_256	RFC 4868 [i.56]				
	ENCR_AES_CBC	RFC 3602 [i.52]				IKEv2 Encryption Algorithm Transforms (RFC 8247 – [i.62])
	ENCR_CHACHA20_POLY1305	RFC 7634 [i.60]				
	ENCR_ASES_GCM_16	RFC 4106 [i.53]				
	ENCR_AES_CCM_8	RFC 4309 [i.54]				

Security Protocol	Traditional Algorithm	Algorithm Development Organisation and document(s)	Algorithm used for	O-RAN Adoption (Interfaces, APIs)	O-RAN specification(s)	
	AUTH_HMAC_SHA2_256_128	RFC 4868 [i.56]	IKEv2 Integrity Algorithm Transforms (RFC 8247 – [i.62])			
	AUTH_HMAC_SHA2_512_256	RFC 4868 [i.56]				
	AUTH_HMAC_SHA1_96	RFC 2404 [i.49], RFC 7296 [i.59]				
	AUTH_AES_XCBC_96	RFC 3566 [i.51], RFC 7296 [i.59]	IKEv2 Pseudorandom Function Transforms (RFC 8247 – [i.62])			
	PRF_HMAC_SHA2_256	RFC 4868 [i.56]				
	PRF_HMAC_SHA2_512	RFC 4868 [i.56]				
	PRF_HMAC_SHA1	RFC 2404 [i.49]				
	PRF_AES128_XCBC	RFC 4344 [i.18]				
	DH 14 – 2048-bit MODP Group	RFC 3526 [i.50]	IKEv2 Diffie-Hellman Group Transforms (RFC 8247 – [i.62])			
	DH 19 – 256-bit random ECP group	RFC 5114 [i.57]				
JWS (NOTE 5)	HS256 (HMAC using SHA-256)	RFC 7518 [i.36]	Message authentication codes	A1 [i.24] R1 [i.25] Y1 [i.26] Near-RT RIC APIs [i.37] AALI-C-Mgmt interface [i.38] O2 [i.39]	SPS [i.14]	
	HS384 (HMAC using SHA-384)	RFC 7518 [i.36]				Signature algorithms
	HS512 (HMAC using SHA-512)					
	RS256 (RSASSA-PKCS1-v1_5 using SHA-256)					
	RS384 (RSASSA-PKCS1-v1_5 using SHA-384)					
	RS512 (RSASSA-PKCS1-v1_5 using SHA-512)					
	ES256 (ECDSA using P-256 curve and SHA-256)					
	ES384 (ECDSA using P-384 curve and SHA-384)					
	ES512 (ECDSA using P-512 curve and SHA-512)					
	PS256 (RSASSA-PSS using SHA-256 and MGF1 with SHA-256)					
	PS384 (RSASSA-PSS using SHA-384 and MGF1 with SHA-384)					
	PS512 (RSASSA-PSS using SHA-512 and MGF1 with SHA-512)					
	JWE	A128CBC-HS256 (AES_128_CBC_HMAC_SHA_256)	RFC 7518 [i.36] RFC 7516 [i.40], clause 4.1.2.			Encryption algorithms
A192CBC-HS384 (AES_192_CBC_HMAC_SHA_384)						
A256CBC-HS512 (AES_256_CBC_HMAC_SHA_512)						
A128GCM (AES GCM using 128-bit key)						
A192GCM (AES GCM using 192-bit key)						
A256GCM (AES GCM using 256-bit key)						
OAuth 2.0		(NOTE 6)		RFC 6749 [i.66]	Authorization framework	
	GCM-AES-128					SPS [i.14]

Security Protocol	Traditional Algorithm	Algorithm Development Organisation and document(s)	Algorithm used for	O-RAN Adoption (Interfaces, APIs)	O-RAN specification(s)
MACsec (NOTE 4)	GCM-AES-256	IEEE 802.1AE-2018 [i.41] IEEE 802.1X-2020 [i.42]	Symmetric encryption and decryption, Message authentication on codes	Open FH CUS Plane Security [i.43] Open FH MPS [i.12]	Open FH MPS [i.12]
	GCM-AES-XPN-128				
	GCM-AES-XPN-256				
	AES-CMAC	RFC 4493 [i.44]	Key derivation		
PTP	HS256 (HMAC using SHA-256)	IEEE 1588 [i.47], clause 16.14	Message authentication on codes	Open FH CUS Plane Security [i.43] Open FH MPS [i.12]	SPS [i.14]
	HS384 (HMAC using SHA-384)				
	HS512 (HMAC using SHA-512)				
NOTE 1: ETM (Encrypt-then-MAC) is not specified for SSH yet is a widely used and recommended practice within the SSHv2 protocol, and broadly supported in available implementations, due to its security- related advantages.					
NOTE 2: BSI recommends phasing out by 2025 [i.33].					
NOTE 3: The related authenticated encryption algorithm is defined in [i.34]					
NOTE 4: EAP-TLS is based on TLS 1.2 (see IETF RFC 5216 [i.45]) or TLS 1.3 (see IETF RFC 9190 [i.46])					
NOTE 5: JWT is based on JWS and JWE (see IETF RFC 7519 [i.65])					
NOTE 6: OAuth 2.0 is based on TLS, JWT, and JWE.					
NOTE 7: AUTH_NONE MUST be supported as per RFCs 7296 and RFC 5282 and MUST NOT be supported when ESP uses Authenticated Encryption with Associated Data (AEAD) cipher.					
NOTE 8: AUTH_AES_XCBC_96 SHOULD be supported as per RFC 3566 and RFC 7296, and MAY be support in case of Interoperability with IoT.					

Annex:

Change history/Change request (history)

Date	Revision	Description
2025.02.04	00.01	First draft
2025.02.25	00.02	Merged NOK-0244, NOK-0245
2025.03.04	00.03	Merged NOK-0256
2025.04.01	00.04	Merged NOK-0268
2025.04.14	00.05	Merged SYM-0084, NIST.AO-0012, NIST-0013
2025.05.27	00.06	Merged NEC-0056
2025.06.10	00.07	Merged NIST-0018, NIST-0019
2025.06.23	00.08	Adopted changes on NIST-0018 and NIST-0019 due to the ongoing review
2025.07.9	01.00	Published as Final version 01.00